📊 Power BI DAX

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📄 Overview

DAX concepts and formulas using Superstore dataset including columns and measures.

🟢 PART 1: Navigation Steps (Columns)

🔹 Step 1: Load Data

1. Open Power BI Desktop
2. Click **Home** → **Get Data** → **Excel**
3. Select dataset
4. Click **Load**

🔹 Step 2: Go to Data View

1. Click **Table Icon (left panel)**
2. View rows and columns

🔹 Step 3: Create New Column

1. Select table (Orders)
2. Click **Modeling** → **New Column**
3. Enter DAX formula
4. Press **Enter**

🔹 Step 4: Verify Output

1. Scroll right
2. Check new column
3. Validate values

🟢 PART 2: Calculated Columns (Row-Level)

✅ Question 1: Total Value

```
Total Value = Orders[Sales] + Orders[Profit]
```

✅ Question 2: Remaining Amount

```
Remaining Amount = Orders[Sales] - Orders[Discount]
```

✅ Question 3: Sales per Quantity

```
Sales per Quantity = Orders[Sales] * Orders[Quantity]
```

✅ Question 4: Profit Status

```
Profit Status = IF(Orders[Profit] > 0, "Profit", "Loss")
```

✅ Question 5: Sales Level

```
Sales Level = IF(Orders[Sales] > 500, "High", "Low")
```

✅ Question 6: Net Sales

```
Net Sales = Orders[Sales] * (1 - Orders[Discount])
```

✅ Question 7: Profit Category

```
Profit Category =
IF(
    Orders[Profit] < 0, "Loss",
    IF(Orders[Profit] <= 100, "Low Profit", "High Profit")
)
```

✅ Question 8: Order Year

```
Order Year = YEAR(Orders[Order Date])
```

✅ Question 9: Sales Category

```
Sales Category =
IF(
    Orders[Sales] < 100, "Low",
    IF(Orders[Sales] <= 500, "Medium", "High")
)
```

✅ Question 10: Month Name

```
Month Name = FORMAT(Orders[Order Date], "MMMM")
```

✅ Question 11: Discount Category

```
Discount Category =
IF(
    Orders[Discount] = 0, "No Discount",
    IF(Orders[Discount] < 0.2, "Low", "High")
)
```

📊 Power BI DAX Measures

🟢 PART 3: Navigation Steps (Measures)

🔹 Step 1: Go to Report View

1. Click **Report Icon (left panel)**

🔹 Step 2: Create Measure

1. Select table
2. Click **Modeling** → **New Measure**
3. Enter DAX
4. Press **Enter**

🔹 Step 3: Use in Visual

1. Insert visual (Table / Bar / Card)
2. Drag measure to **Values**

🟢 PART 4: Basic Measures

✅ Question 1: Total Sales

```
Total Sales = SUM(Orders[Sales])
```

✅ Question 2: Total Profit

```
Total Profit = SUM(Orders[Profit])
```

✅ Question 3: Total Orders

```
Total Orders = COUNT(Orders[Order ID])
```

✅ Question 4: Average Sales

```
Average Sales = AVERAGE(Orders[Sales])
```

✅ Question 5: Max Sales

```
Max Sales = MAX(Orders[Sales])
```

✅ Question 6: Min Sales

```
Min Sales = MIN(Orders[Sales])
```

🟢 PART 5: Conditional Measures

✅ Question 7: Positive Profit

```
Positive Profit =
CALCULATE(
    SUM(Orders[Profit]),
    Orders[Profit] > 0
)
```

✅ Question 8: High Sales Total

```
High Sales Total =
CALCULATE(
    SUM(Orders[Sales]),
    Orders[Sales] > 500
)
```

✅ Question 9: Orders with Discount

```
Orders with Discount =
CALCULATE(
    COUNT(Orders[Order ID]),
    Orders[Discount] > 0
)
```

🟢 PART 6: Intermediate Measures

✅ Question 10: Profit Ratio

```
Profit Ratio = DIVIDE(SUM(Orders[Profit]), SUM(Orders[Sales]))
```

✅ Question 11: Total Discount

```
Total Discount =
SUMX(
    Orders,
    Orders[Sales] * Orders[Discount]
)
```

✅ Question 12: Average Profit per Order

```
Avg Profit per Order =
DIVIDE(
    SUM(Orders[Profit]),
    COUNT(Orders[Order ID])
)
```

✅ Question 13: Sales per Customer

```
Sales per Customer =
DIVIDE(
    SUM(Orders[Sales]),
    DISTINCTCOUNT(Orders[Customer ID])
)
```

🟢 PART 7: Difference (Column vs Measure)

Feature	Column	Measure
Level	Row-level	Aggregated
Stored	Yes	No
Use	Data preparation	Visualization

📊 DAX Formulas by Category (Power BI)

🟢 1. Aggregation Functions

```
Total Sales = SUM(Orders[Sales])

Total Profit = SUM(Orders[Profit])

Average Sales = AVERAGE(Orders[Sales])

Total Orders = COUNT(Orders[Order ID])

Unique Customers = DISTINCTCOUNT(Orders[Customer ID])

Max Sales = MAX(Orders[Sales])

Min Sales = MIN(Orders[Sales])
```

🟢 2. Logical Functions

```
Profit Status = IF(Orders[Profit] > 0, "Profit", "Loss")

Sales Level = IF(Orders[Sales] > 500, "High", "Low")

Profit Category =
IF(
    Orders[Profit] < 0, "Loss",
    IF(Orders[Profit] <= 100, "Low Profit", "High Profit")
)
```

🟢 3. Mathematical Functions

```
Total Value = Orders[Sales] + Orders[Profit]

Remaining Amount = Orders[Sales] - Orders[Discount]

Sales per Quantity = Orders[Sales] * Orders[Quantity]

Profit Ratio = DIVIDE(SUM(Orders[Profit]), SUM(Orders[Sales]))
```

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4. Filter Functions

```
High Sales Total =
CALCULATE(
    SUM(Orders[Sales]),
    Orders[Sales] > 500
)

Positive Profit =
CALCULATE(
    SUM(Orders[Profit]),
    Orders[Profit] > 0
)

Orders with Discount =
CALCULATE(
    COUNT(Orders[Order ID]),
    Orders[Discount] > 0
)
```

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5. Iterator Functions

```
Total Discount =
SUMX(
    Orders,
    Orders[Sales] * Orders[Discount]
)

Average Sales per Order =
AVERAGEX(
    Orders,
    Orders[Sales]
)
```

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6. Date & Time Functions

```
Order Year = YEAR(Orders[Order Date])

Month Name = FORMAT(Orders[Order Date], "MMMM")
```

```
Month Number = MONTH(Orders[Order Date])

Day Name = FORMAT(Orders[Order Date], "dddd")
```

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7. Text Functions

```
Customer Full Name = Orders[Customer Name] & " - " & Orders[Segment]

Upper Region = UPPER(Orders[Region])

Lower Region = LOWER(Orders[Region])
```

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8. Relationship Functions

```
Dept Name = RELATED(Dept[DeptName])

Employees per Dept = COUNTROWS(RELATEDTABLE(Emp))
```

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9. Ranking Functions

```
Sales Rank =
RANKX(
    ALL(Orders),
    SUM(Orders[Sales]),
    ,
    DESC
)
```

---

10. Time Intelligence Functions

```
Sales Last Year =
CALCULATE(
    SUM(Orders[Sales]),
    SAMEPERIODLASTYEAR(Orders[Order Date])
)

Year to Date Sales =
TOTALYTD(
    SUM(Orders[Sales]),
    Orders[Order Date]
)
```

In [ ]: